Roomify — technical report

Redesign more, buy less. Devpost: https://devpost.com/software/roomify-hsc4yn · Code: backend · CV service · frontend

> **Draft — release candidate v0.9.0-rc1 (2026-10-24).** Everything below is implemented, merged and tested. AI quality, latency, cost and budget-compliance results (§6.2) are **pending: requires OpenAI key** — no key was available, so they are left empty rather than estimated, and the AI features have not yet been run against the live API.

1. Problem

People who most want to improve their room — students, renters, young adults on small budgets — are the least served by existing tools (scope):

1. **They can't picture a redesign in their own room.** Inspiration sites show other people's spaces.
2. **They can't see the total cost**, or what to cut when a plan is over budget.
3. **They often can't drill, paint or replace flooring**, yet most ideas assume they can.
4. **Usable furniture gets replaced** instead of kept, moved, repaired or upcycled.

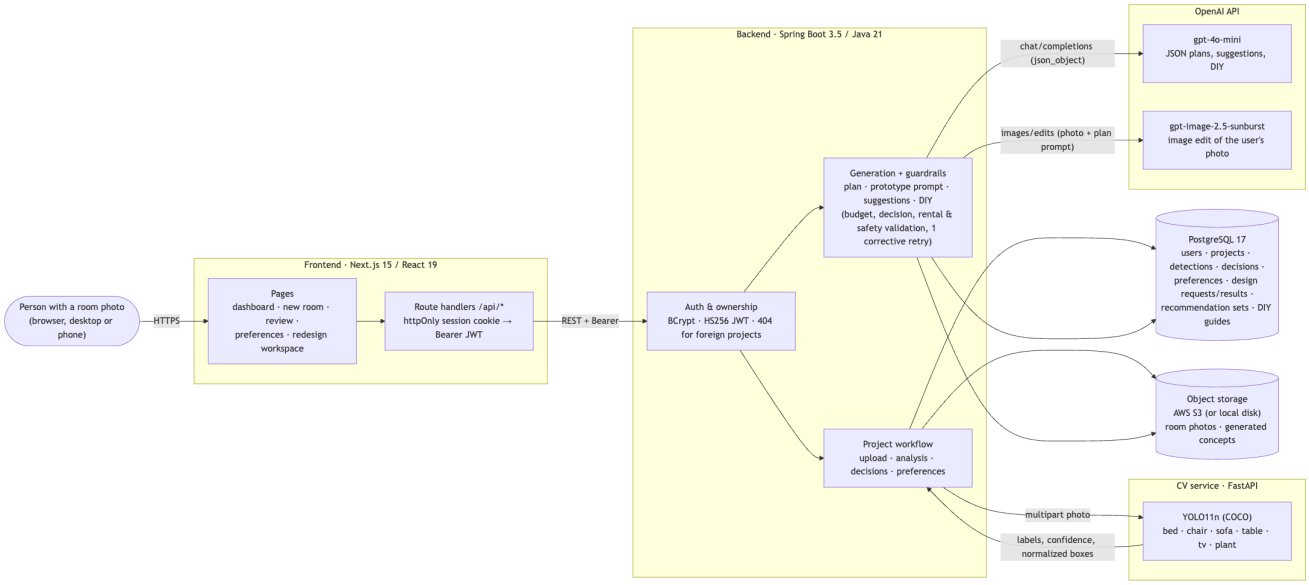
2. Approach

One photo of the real room becomes one organised, saved project:

1. **Detect** the major furniture in the photo (computer vision).
2. **Let the person correct and decide**: Keep, Replace, Remove or Unsure per item; rename, remove or add items the detector got wrong. Detection is treated as a draft, not ground truth.
3. **Collect constraints**: style, colours, room purpose, maximum budget, Rental-Friendly Mode (on by default for renters and dorms), free-text requirements.
4. **Generate a plan with an LLM, then verify it in code**: every kept item is kept, every removed item removed, the costs add up and fit the budget, rental mode contains no permanent work. A plan that breaks a rule is retried once with the reason, otherwise rejected — never shown.
5. **Render a visual concept** by editing the person's own photo with an image model, guided by the verified plan.
6. **Suggest purchases and DIY** within the budget: estimated price, placement, reason, a cheaper alternative, and beginner DIY guides that pass a safety filter.
7. **Save everything** under the person's account; the dashboard reopens any project where it stopped.

The design principle is *the model proposes, the server disposes*: LLM output is untrusted input that must pass deterministic validation before it is stored.

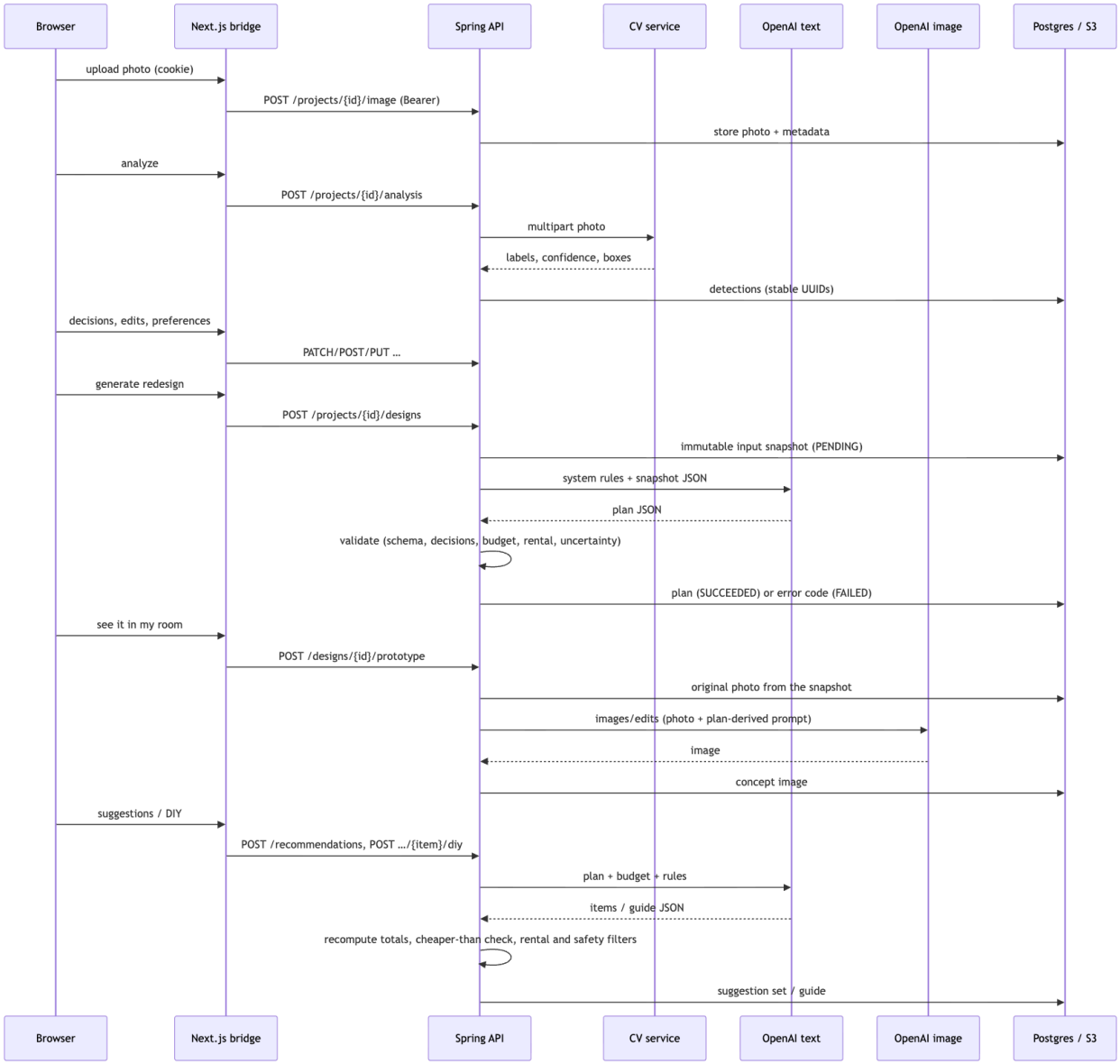
3. System design



Source: `architecture/architecture.mmd`.

Service	Stack	Responsibility
Frontend	Next.js 15.5, React 19, TypeScript, Tailwind	Pages; a same-origin bridge (`/api/*` route handlers) that keeps the JWT in an httpOnly cookie and forwards it as a Bearer token
Backend	Spring Boot 3.5, Java 21, PostgreSQL 17 + Flyway	Auth and ownership, workflow state, storage, provider calls, all validation
CV service	FastAPI, Ultralytics YOLO11n	Furniture detection; private, called only by the backend
Storage	AWS S3 (or local disk)	Room photos and generated concepts
AI	OpenAI	Plans, suggestions, DIY (text); concept image (image edits)

3.1 Data flow



Source: `architecture/data-flow.mmd`.

3.2 Data model (main tables)

`users` → `room_projects` (owner, name, room type, rental status, workflow status) → `room_images` → `detected_objects` (stable UUID, source CV/MANUAL, active flag) → `furniture_decisions`; `project_preferences`; `design_requests` (immutable input snapshot, provider, prompt version) → `design_results` (plan JSON, status, error code) → `design_assets` (concept image); `recommendation_sets` → `recommendation_items` → `recommendation_diy_guides`. Ten Flyway migrations.

4. Models and providers

Task	Model	Why	Settings
Furniture detection	Ultralytics **YOLO11n**, COCO-pretrained	Runs on CPU in ~32 ms, 5 MB, no training needed	confidence ≥ 0.25; six classes kept: bed, chair, sofa, table, tv, plant
Plan, suggestions, DIY	OpenAI **gpt-4o-mini** (Chat Completions, `json_object`)	Cheapest model that follows long JSON schemas	temperature 0.1, ≤ 4000 output tokens, 1 corrective retry
Visual concept	OpenAI **gpt-image-2.5-sunburst** (`images/edits`)	Edits the user's real photo; OpenAI's replacement for the retired `gpt-image-1`, positioned for precise edits (not yet called live)	1024×1024, quality `low`

Model names are configuration. Two provider facts discovered during the build changed the defaults: `gpt-image-1` was shut down by OpenAI on 2026-10-23, and `gpt-4o-mini` shuts down on 2027-01-14 (its replacement `gpt-6-luna` is a reasoning model; the request code now supports it by configuration).

5. Key decisions

1. **Detection is a draft.** COCO has no desk, wardrobe, shelf or lamp class, and measured recall is 0.62 (§6.1). So the review step supports rename, remove and add, low-confidence items are flagged, and analysis failure offers manual entry instead of a dead end.
2. **Validate generated output in code.** Plans must contain every detected item exactly once with its saved decision; instructions must match the decision ("Keep …", "Replace …"); allocations must sum exactly to the total and fit the budget (BigDecimal); rental plans are scanned for permanent work; low-confidence and Unsure items need uncertainty wording. Suggestions get server-computed totals and strictly-cheaper alternatives; DIY guides are Beginner/Easy, cheaper than buying, and rejected if they mention electrical, plumbing, gas or structural work.
3. **Immutable input snapshots.** Each generation stores exactly what the model saw (detections, decisions, preferences, image reference). The concept is rendered from the snapshot's photo, so it always matches its plan, and a design whose inputs changed mid-generation is rejected.
4. **Privacy by construction.** One authorization rule covers every `/api/projects/{id}/**` route; another user's project is indistinguishable from a missing one (404). The browser never holds the JWT. The CV service is designed to stay off the public internet, and its `image_url` input refuses non-public addresses (SSRF).
5. **Estimates, said plainly.** Prices, totals and the concept image are labelled as estimates and concepts everywhere they appear; the plan page states it is not a professional plan.
6. **Budget warnings that stay true.** Stored suggestions always fit the budget at generation time; if the person lowers the budget later, the summary warns and shows what switching to the cheaper options would save.

6. Results

6.1 Detection (measured)

60 COCO val2017 indoor images (held out from training), 159 objects, matched at IoU ≥ 0.5 through the product's HTTP API. Full table, method and failure cases: EVALUATION.md.

	Precision	Recall	Recall, large objects
All six classes	**0.70**	**0.62**	0.68
Best (bed)	0.86	0.74	0.73
Worst (table)	0.40	0.55	0.67

Latency: 32 ms p50 / 41 ms p95 model time on a laptop CPU.

6.2 Generation

Result	Value
Plan budget compliance	**Pending: requires OpenAI key**
Plan decision and rental compliance	**Pending: requires OpenAI key**
Suggestions within budget	**Pending: requires OpenAI key**
DIY safety-filter rejections	**Pending: requires OpenAI key**
Latency per AI step	**Pending: requires OpenAI key**
Cost per redesign	**Pending: requires OpenAI key**

Method: EVALUATION.md §2. Verified without a key: every AI action fails with a clear `AI_CONFIGURATION_ERROR` message and leaves the project usable (EVALUATION.md §3).

6.3 Engineering quality

Suite	Result (develop, 2026-10-24)
Backend JUnit (Postgres 17.11)	149 passed
CV pytest	21 passed
Frontend Vitest	230 passed
Playwright, stubbed API, chromium + mobile	39 passed, 4 consecutive runs
Playwright, fake Spring API through Next	6 passed
Playwright, **real stack** (no AI key)	full journey, privacy, logout, CV-down fallback passed

CI runs on every push and PR in all three repositories. Secret scan of all history (including unreachable objects): no secrets.

7. Limitations

- **Detection** misses about a third of objects, most of them small; it cannot detect desks, wardrobes, shelves, lamps or rugs at all (COCO classes). Measured on 60 COCO images, not on user photos.
- **No spatial understanding.** One photo gives no scale; the plan never claims dimensions or fit.
- **Concepts are not photos.** The image model may change proportions, windows or lighting and does not reproduce kept furniture exactly.
- **Prices are model estimates**, not retailer data; there is no live shopping API.
- **Rental and safety checks are keyword-based** and cannot certify a lease or a DIY step as safe.
- **One room, one photo** per project (as scoped).

- **Provider dependence.** Two of three models changed status during the build; defaults may need updating again. The text model's deprecation date (2027-01-14) is documented.
- **AI features unverified live.** Plan, concept, suggestion and DIY generation have only run against fake providers; the default image model (`gpt-image-2.5-sunburst`) has never been called.

8. Future work

- Fine-tune a detector on interior datasets with desk, wardrobe, shelf, lamp and rug classes.
- Multiple photos per room for better coverage; optional measurements for fit checks.
- Real product search with live prices and second-hand listings.
- Before/after slider, several concept variations, and a follow-up chat ("make it cosier", "cut cost").
- Run generation asynchronously with progress updates instead of one long request.

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